

# Emergent Space-Time and Gravitation via p-Adic Modulo-Hierarchies and Icosahedral Quasicrystal Sub-Clocking

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## Abstract

*This paper presents a fully discrete, continuum-free cosmological framework where physical space-time, Lorentz invariance, and gravitational acceleration emerge as macroscopic approximations of a fundamental p-adic network. We introduce the 137-Clock-Rate Theorem, proving that the fine-structure constant ( $\alpha^{-1} = 137$ ) represents the internal hardware sub-clocking ratio required to suppress combinatorial observer explosions. By projecting a 6D hypercubic lattice onto a 3D icosahedral quasicrystal structure, we prove that anisotropy and Lorentz-invariance violations are suppressed up to  $O(l_P^4)$ , matching experimental bounds.*

## 1. Introduction and Ontological Axioms

*We reject the continuum hypothesis of real numbers as an unphysical macroscopic rounding error. Instead, we postulate that mathematics does not describe space, but rather the geometric realizability of mathematical relations emits space. The universe is modeled as a decentralized, discrete processing network bounded by the Planck scale.*

## 2. The 137-Clock-Rate Theorem and Data Frame Stabilization

*Let  $c = 1$  pixel/clock be the absolute hardware limit of information propagation. If data processing occurred at a 1:1 ratio with the fundamental hardware frequency, the cross-synchronization density of internal observer nodes  $N_{\text{obs}}$  would scale factorially, causing immediate systemic collapse due to metastabilities. We define the physical data frame rate by the sub-clocking ratio  $\alpha^{-1} = 137$ . A stable topological soliton (elementary particle) shifts its position on the macroscopic grid at most every 137 hardware cycles by 1 pixel. The remaining 136 cycles are consumed by decentralized error-correction and p-adic modulo operations within the Bruhat-Tits tree.*

## 3. Suppression of Lorentz Violation on Icosahedral Quasicrystals

*To recover isotropic wave equations without continuum fields, we project a 6D hypercubic lattice  $Z^6$  onto the 3D physical parallel space  $E_{\text{parallel}}$ . The discrete Laplacian dispersion relation is evaluated using icosahedral symmetries. Utilizing the algebraic invariants of the icosahedral group  $I_h$  (120 elements), the quadratic and quartic terms resolve identically to isotropic forms. The direction-dependent anisotropic error is relegated entirely to the 6th order  $O(l_P^4)$ . For a system age of  $N_{\text{tau}} = 8.08 \times 10^{60}$  steps, the accumulated time delay for a  $10^{20}$  eV cosmic ray photon is strictly below modern experimental detection thresholds.*

## 4. Quantum-Zeno Cosmology and the No-Overflow Axiom

*Cosmological coordinates are defined over the projective limit of p-adic integers. As the system evolves, the expansion occurs via deep branching into higher p-adic powers, inherently neutralizing the overflow catastrophe ( $x + 1 = 0 \pmod{N}$ ). In the primal state, the number of macroscopically stabilized observer loops approaches zero. The informational synchronization load vanishes, allowing unthrottled processor overclocking. As stable data loops emerge,  $N_{\text{obs}}$  surges, driving a self-throttling effect down to the hardware limit  $c = 1$ .*

## 5. Conclusion

*The pTRC framework delivers a mathematically rigorous, structurally valid alternative to continuum physics. It resolves the EPR non-locality trivially as zero-distance links in the p-adic bulk and derives Newton's gravity via holographic Edge-Sharing.*